

Introduction to Cybersecurity

Cybersecurity is the practice of protecting **systems, networks, applications, cloud infrastructure, digital data, and identities** from unauthorized access, malware, ransomware, data theft, and AI-enabled attacks. Every connected device and organization is a potential target.



CORE CONCEPT

The CIA Triad: Three Pillars of Information Security

Every cybersecurity decision maps back to three foundational principles.

Confidentiality

Ensure only authorized users access sensitive data.

Integrity

Keep data accurate and unaltered without authorization.

Availability

Keep systems and data accessible when needed.



When Cybersecurity Fails



Financial Loss

Ransom payments, lost revenue, recovery costs, and regulatory fines can total millions.



Operational Disruption

Hospitals cancel procedures. Airlines ground flights. Power grids and water systems go offline.



AI-Powered Threats

AI writes novel exploits, automates vulnerability scanning, and generates convincing phishing emails and deepfakes.

Real-world example: In 2021, a hacker remotely accessed a Florida water treatment facility and attempted to poison the water supply. [Read the full story.](#)



Real-World Incident: Complete Attack Lifecycle



Firewall Access

Compromised admin credentials used from suspicious IP 23.234.107.253.



Config Exfiltration

Firewall config file stolen — revealing network topology and VPN details.



SSL VPN Compromise

Second admin account used to authenticate to SSL VPN using exfiltrated data.



Internal Network Access

Perimeter defenses bypassed — attacker now has a foothold inside the network.

Persistence to Payload

1. Persistence - SMBExec

```
%COMSPEC% /Q /c cmd.exe  
/Q /c for /f "tokens=1,2  
delims= " ^%A in ("tasklist /fi  
"Imagename eq lsass.exe" |  
find "lsass""") do rundll32.exe  
C:\windows\System32\comsvc  
s.dll, #+0000^24 ^%B  
\Windows\Temp\OM7c.iso full
```

2. Lateral Movement

```
C:\Users\ADMINI~1\AppData  
a\Local\Temp\3582-  
490\mimikatz.exe
```

Mimikatz extracts plaintext passwords and NTLM hashes from memory, granting **domain-wide access** through pass-the-hash attacks.

3. Ransomware

Deploys INC ransomware across the network, dropping **INC-README.txt** on every system — encrypting files while threatening to leak exfiltrated data (**double extortion**).

Ransom Note

```

~~~~~ INC Ransom ~~~~~

-----> Your data is stolen and encrypted.
If you don't pay the ransom, the data will be published on our TOR darknet sites.
The sooner you pay the ransom, the sooner your company will be safe.

Tor Browser Link:
http://incblog6qu4y4mm4zvwm5nrmue6qbwgtgjsxpwb67ixzssu36tsajldoad.onion/

Link for normal browser:
http://incblog.su/

-----> What guarantees are that we won't fool you?
We are not a politically motivated group and we want nothing more than money.
If you pay, we will provide you with decryption software and destroy the stolen data.
After you pay the ransom, you will quickly restore your systems and make even more money.
Treat this situation simply as a paid training for your system administrators, because it is due
to your corporate network not being properly configured that we were able to attack you.
Our pentest services should be paid just like you pay the salaries of your system administrators.
Get over it and pay for it.
If we don't give you a decryptor or delete your data after you pay, no one will pay us in the
future.
You can get more information about us on Twitterhttps://twitter.com/hashtag/incransom?f=live

-----> You need to contact us on TOR darknet sites with your personal ID
Download and install Tor Browserhttps://www.torproject.org/
Write to the chat room and wait for an answer, we'll guarantee a response from you.
Sometimes you will have to wait some time for our reply, this is because we have a lot of work and
we attack tens of companies around the world.

Tor Browser Link for chat:
http://incpaykabjgc2mtdxq6c23nqh4x6m5dkps5fr6vgdkgzb5njsxx6qkid.onion/

Your personal ID:
6a10053cd15*****

-----> Warning! Don't delete or modify encrypted files, it will lead to problems with decryption
of files!
-----> Don't go to the police or the FBI for help. They won't help you.
The police will try to prohibit you from paying the ransom in any way.
The first thing they will tell you is that there's no guarantee to decrypt your files and remove
stolen files.

```



Major Cybersecurity Domains



Security Operations Center

Monitor, triage, and respond to threats in real time.



Incident Response & DFIR

Investigate breaches, determine root cause, and lead recovery.



Penetration Testing

Simulate attacks to find and fix vulnerabilities before adversaries do.



Governance, Risk & Compliance

Manage risk, enforce policies, and meet regulatory requirements.

Security Operations Center (SOC)



Core Responsibilities

- Monitor and triage security alerts 24/7
- Investigate and contain active incidents
- Run proactive threat hunts
- Coordinate incident response

Common Tools

SIEM: Splunk, Microsoft Sentinel, IBM QRadar

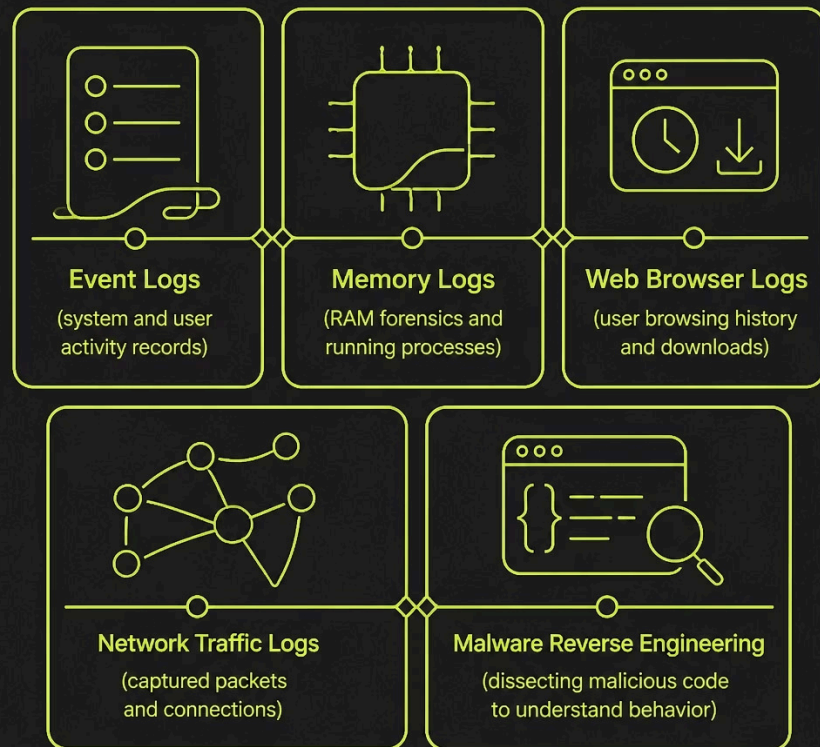
EDR: CrowdStrike, Microsoft Defender, SentinelOne

AI's Role

Summarizes alerts, correlates events, reduces false positives, and cuts alert fatigue.

Incident Response & Digital Forensics (DFIR)

DFIR goes beyond the SOC — determining **what happened, how, and what was impacted** to guide remediation and recovery.



What DFIR Uncovers

- Attack timeline and entry point
- Scope of data accessed or stolen
- Malware behavior and persistence mechanisms
- Evidence for legal proceedings

Penetration Testing & Governance, Risk and Compliance

Penetration Testing

Finding vulnerabilities before attackers do.

- **Web App Testing:** OWASP Top 10
- **Network Testing:** Open ports, weak passwords, vulnerable services, misconfigurations
- **Tools:** Kali Linux, Nmap, Metasploit, Burp Suite

Governance, Risk & Compliance

Managing risk and enforcing security policies.

- **Risk Assessment:** What can go wrong? Impact? Likelihood?
- **Policies:** Password rules, acceptable use, data retention, incident response

Skills That Power Every Cybersecurity Domain

Networking

TCP/IP, DNS, HTTP/HTTPS, firewalls, VPNs, ports & protocols

Operating Systems

Windows (Active Directory, Event Logs, PowerShell) & Linux (CLI, Bash, SSH)

Scripting

Python, PowerShell, and Bash for automation and analysis

Security Fundamentals

CIA Triad, Defense in Depth, Least Privilege, Zero Trust, Risk Management

Communication

Explain technical issues to non-technical audiences, write reports, decide under pressure

AI Adaptability

Leverage AI for threat detection, analysis, and automation

Key Certifications for a Cybersecurity Career

Certifications validate specialized skills and demonstrate commitment to continuous learning, making them essential for career advancement in cybersecurity.

Foundational & Entry-Level



CompTIA Network+

Establishes essential networking concepts.



CompTIA Security+

Covers core security principles and best practices.



ISC2 CC

Introduces foundational cybersecurity concepts and career paths.



Google Cybersecurity Certificate

Entry-level pathway to a cybersecurity role.

Penetration Testing (OffSec)



eLearnSecurity Junior Penetration Tester

Practical, hands-on penetration testing skills for beginners.



Practical Network Penetration Tester

Focuses on real-world network penetration testing scenarios.

Defensive & Incident Response



Security Blue Team Level 1 (BTL1)

Practical skills for defensive security, threat hunting, and incident response.



GCIH (GIAC Certified Incident Handler)

Advanced skills in incident handling and response techniques.

Career Paths



AI and the Future of Cybersecurity

● AI is Helping Attackers

- Hyper-personalized phishing
- Deepfake impersonation
- Malware obfuscation
- Automated reconnaissance

● AI is Helping Defenders

- Threat detection
- Incident investigations
- Threat intelligence
- Security copilots

□ AI is making cyberattacks smarter, but it is also empowering defenders with faster and more accurate security capabilities.

Emerging Domain: AI Security



What Needs Protection?

- AI Models
- AI Agents
- Training Data
- LLM Applications



New Threats

- Prompt Injection
- Data Poisoning
- Model Theft
- Sensitive Data Leakage



Emerging Careers

- AI Security Engineer
- AI Red Teamer
- AI Governance Specialist
- LLM Security Researcher

❏ Securing AI is becoming one of the fastest-growing areas in cybersecurity, creating new challenges and career opportunities.

Your Cyber-Security Journey Starts Here

The digital world needs defenders. That defender could be you.

What It Takes to Succeed

Why Cybersecurity Matters

Every organization, from hospitals to governments, depends on security professionals to protect critical data and infrastructure. The demand has never been higher — and it's still growing.

The Role of AI

AI is transforming cybersecurity — both as a tool for defenders and a weapon for attackers. Understanding AI-powered threats and defenses is no longer optional; it's a career essential.

Your Path Forward

Start with foundational certifications, build hands-on skills, and never stop learning. Curiosity, resilience, and continuous growth are the true hallmarks of a great cybersecurity professional.

Build Your Own Active Directory Environment

Active Directory (AD) is Microsoft's directory service used by **95% of Fortune 500 companies** to manage users, computers, and resources across a network. For cybersecurity professionals, AD is ground zero — it's the most targeted system in enterprise attacks, and understanding how to build, secure, and attack it is a **core career skill**.

 This lab gives you safe, hands-on experience with the same infrastructure used in real-world enterprise environments — and exploited in real-world breaches.

95%

Enterprise Adoption

Fortune 500 companies use Active Directory

#1

Attack Target

Most targeted system in enterprise intrusions



Step-by-Step Lab Setup

1

System Requirements

- **Hypervisor:** VirtualBox (free) or VMware Workstation
- **VM 1:** Windows Server 2019/2022 (Domain Controller) — 4GB RAM, 60GB disk
- **VM 2:** Windows 10/11 (Client Machine) — 2GB RAM, 40GB disk
- **Host Machine:** Minimum 16GB RAM recommended
- Set both VMs to the same **Host-Only or Internal Network**

2

Install & Configure Domain Controller

- Install Windows Server and set a static IP (e.g., 192.168.1.10)
- Open **Server Manager** → **Add Roles** → select **Active Directory Domain Services**
- Promote server to Domain Controller
- Create a new forest, e.g., lab.local
- Set DSRM password and complete installation — server will reboot

3

Create Users, Groups & OUs

- Open **Active Directory Users and Computers (ADUC)**
- Create Organizational Units: IT, HR, Finance
- Add user accounts inside each OU (e.g., jsmith, alopez)
- Create Security Groups: Domain Admins, IT-Staff, HR-Staff
- Assign users to appropriate groups

4

Join Client Machine to Domain

- On Windows 10/11 VM, set DNS to Domain Controller IP (192.168.1.10)
- Go to **Settings** → **System** → **About** → **Advanced System Settings**
- Click **Change** under Computer Name, select **Domain**
- Enter lab.local and provide Domain Admin credentials
- Restart the client — it is now domain-joined!

5

Test Auth & Group Policies

- Log into the client as a domain user (e.g., lab\jsmith)
- Open **Group Policy Management Console (GPMC)** on the DC
- Create a GPO to enforce a password policy or desktop wallpaper
- Run gpupdate /force on the client to apply policies
- Verify the policy is applied with gpresult /r

Active Directory Lab: Key Concepts & Outcomes

Key Concepts to Explore



NTLM & Kerberos

Understand how domain authentication works. Use tools like Wireshark to capture and inspect Kerberos tickets (TGT/TGS) and NTLM challenge-response handshakes.



Privilege Escalation

Explore how attackers abuse misconfigured permissions, weak ACLs, and nested group memberships to escalate from a standard user to Domain Admin.



Group Policy Objects

GPOs control everything from password complexity to software installation. Experiment with GPOs that enforce security baselines and restrict user privileges.



Security Groups

Practice the principle of least privilege. Understand how group memberships control access to resources — and why over-privileged accounts are a critical risk.

Learning Outcomes

After completing this lab, you will be able to:

- Deploy and configure a fully functional Active Directory domain from scratch
- Manage users, groups, and OUs using enterprise-grade tools
- Understand how Kerberos and NTLM authentication protect (and expose) a domain
- ⇒ Apply Group Policy to enforce security controls across the environment
- ↘ Identify common misconfigurations that lead to privilege escalation attacks

Troubleshooting Tips

Can't join the domain?

Ensure the client's DNS is pointing to the DC's IP, not a public DNS like 8.8.8.8. Run `nslookup lab.local` to verify.

GPO not applying?

Run `gpupdate /force` and check `gpresult /h report.html` for a detailed HTML report on applied policies.

Authentication failures?

Check Event Viewer on the DC under **Windows Logs → Security** for Event IDs 4625 (failed login) and 4740 (account lockout).

VM performance issues?

Enable VT-x/AMD-V in your BIOS, allocate sufficient RAM, and use an SSD. Disable unnecessary Windows features on both VMs.